

Technical Document #413: Database Systems - Comprehensive Overview

Technical Document #413: Database Systems - Comprehensive Overview

ACID properties (Atomicity, Consistency, Isolation, Durability) ensure reliable transaction processing in databases.

Sharding distributes data across multiple database servers. It improves performance by parallelizing query processing.

Database migration tools manage schema changes and data transformations. Flyway and Liquibase are popular database migration tools.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Key Takeaways:

- NoSQL databases handle unstructured and semi-structured data.
- Sharding distributes data across multiple database servers.
- Database replication copies data across multiple servers.